

11. Full Development Roadmap — Accelerated Delivery

To deliver the complete DRS platform, all feature tracks run in parallel using a larger, specialized team organized into three concurrent engineering streams. This replaces the sequential phase model with a synchronized parallel delivery model — every stream contributes to the same shared infrastructure while owning its own domain.

Accelerated Delivery Strategy: Three Parallel Engineering Streams

Stream A — Core Platform (Windows Agent + Relay + Command Center):

Stream B — Cross-Platform Agents (macOS + Linux):

Stream C — Mobile MDM (Android + iOS):

Foundation Sprint (All Streams)

All three streams start with shared infrastructure in Month 1

Architecture Lock & Environment Setup

- Finalize complete system architecture — Agent protocol, Relay gateway, Command Center API surface
- Lock full technology stack — Go backend, C++ Windows agent, React frontend, WebRTC protocol
- Provision AWS infrastructure — EC2 auto-scaling, RDS PostgreSQL, ElastiCache Redis, S3 with object lock, CloudFront CDN
- Deploy coturn TURN/STUN relay servers in US-East-1 and US-West-2 — configured and load-tested
- Set up GitHub Actions CI/CD pipeline — automated build, test, and Docker image publish per stream
- Implement core security framework — AES-256, TLS 1.3, RSA-2048 key exchange, JWT authentication
- Design and freeze complete database schema for all modules including MDM tables

Shared Protocol & API Layer

- Implement unified Agent-to-Relay binary protocol (Protobuf over WebSocket) — used by all agent types
- Build multi-tenant authentication service — MFA, RBAC roles, API key management
- Stand up Command Center shell — device list, tenant management, and session launcher scaffolding
- Deploy monitoring stack — Grafana + Prometheus + CloudWatch alerting from Week 1
- Hire all stream B and C engineers during Month 1 so they onboard at Month 2 with architecture already in place

Core Build Sprint (All Streams Parallel)

Heaviest engineering period — all 3 streams building simultaneously

Stream A: Windows Agent + Remote Engine

- Windows Agent service in C++ — silent .msi install, auto-start, tamper protection, <50MB RAM target
- WebRTC screen capture and streaming pipeline — sub-150ms latency with adaptive bitrate compression
- High-concurrency Relay/Gateway in Go — WebSocket session multiplexing, 500+ concurrent sessions
- Unattended (persistent) and Attended (join-via-code) access modes
- Bidirectional drag-and-drop file transfer with SHA-256 integrity validation
- All 6 control modes: View Only, Full Control, Input Lock, Annotate, Parallel Desktop, Backstage
- Remote PowerShell and CMD shell gateway — xterm.js frontend, full history logging
- Live Process Manager, Service Controller, Registry Editor, Event Log Viewer

Stream A: Command Center Portal

- Full React 18 + TypeScript portal — device inventory, real-time online/offline, session launcher
- On-screen alert system — full-screen takeover, banner, popup with industry-specific templates
- Script library with version control, scheduling engine, and bulk deployment across endpoints
- Windows patch management — OS + 200+ third-party application patches
- Automation rule engine — trigger-based policy actions
- Automatic session recording — encrypted S3 storage with tamper-evident object lock
- Immutable audit trail — every action logged with full forensic detail

Stream B: macOS & Linux Agents

- macOS Agent in Swift/Obj-C — ScreenCaptureKit screen capture, Accessibility API input injection
- macOS Fast User Switching parallel desktop via dscl + VNC-over-tunnel
- Bash/Zsh shell gateway, file system browser, process manager via macOS APIs
- macOS patch management — system updates via softwareupdate + Homebrew package manager
- Linux Agent in C/C++ — Ubuntu 20.04+, Debian 10+, RHEL 8+ multi-distro support
- Xvfb virtual framebuffer for parallel desktop on Linux (separate X display :1 for manager)
- systemd service management, apt/yum/dnf package manager patching integration

- Python 3 scripting engine — cross-platform execution on macOS and Linux

Stream C: Android & iOS MDM

- Android Work Profile enrollment — Android Enterprise Device Policy Controller
- MediaProjection API for live screen view on Android 10+
- Real-time GPS tracking and geofencing alert engine — configurable boundary per device
- iOS MDM server implementation — Apple MDM protocol, DEP/ABM enrollment, APNS push integration
- Supervised iOS device management — configuration profiles, certificate deployment, app distribution
- App whitelist/blacklist enforcement — managed Google Play and Apple VPP integration
- Remote wipe — full device wipe and work-profile-only selective wipe (Android + iOS)
- Mobile policy engine — Wi-Fi configs, VPN profiles, camera/mic restrictions, USB control
- MDM portal console — GPS map view, compliance dashboard, geofencing management

Integration & Cross-Platform Unification

All streams merge into unified platform; cross-stream testing begins

- Unified policy engine — one policy set applied across Windows, macOS, Linux, Android, and iOS simultaneously
- Cross-platform patch compliance dashboard — real-time patch status across all OS types in one view
- Consolidated MSP multi-tenant portal — per-client data isolation, per-client branding, per-client policy sets
- Cross-stream automated test suite — end-to-end tests across all agent types against shared relay
- PSA integration foundations — ConnectWise Manage and HaloPSA webhook connectors
- SSO implementation — SAML 2.0 / OIDC with Okta and Azure AD / Entra ID
- Active Directory and Azure AD sync for automatic device enrollment and access lifecycle
- SaaS billing module — Stripe integration, per-endpoint and per-technician subscription management
- Advanced reporting engine — custom report builder, scheduled delivery, executive dashboards
- Performance stress test — 500+ simultaneous endpoints across all agent types

QA, Security, Compliance & US Market Launch

Full platform hardening, external validation, and production launch

Security & Compliance

- Third-party security penetration test across all agent types and relay infrastructure
- Remediation sprint for all critical and high-severity pen test findings
- EV code signing certificate applied to all installers (.msi, .pkg, .deb, .rpm, .apk)
- AV vendor whitelisting submissions — Microsoft Defender, CrowdStrike, Sophos, Malwarebytes
- SOC 2 readiness assessment — activate Drata for automated evidence collection
- Legal review — Terms of Service, Privacy Policy, Data Processing Agreement for US market

Beta & Production Launch

- Beta deployment with 3–5 US-based pilot MSP clients — structured feedback collection
- Performance validation — latency benchmark (sub-150ms), agent footprint (<50MB RAM) verified
- Session recording and audit trail end-to-end validation — zero-gap test over 72-hour period
- All 6 control modes validated across Windows 10/11, Server 2022, macOS 13/14, Ubuntu 22.04
- MDM compliance validated across Android 12/13/14 and iOS 16/17/18
- Production infrastructure hardened — WAF, DDoS protection, failover tested
- Product Hunt launch + US MSP community go-to-market (Reddit r/msp, MSP forums, LinkedIn)
- Full platform declared generally available (GA) —